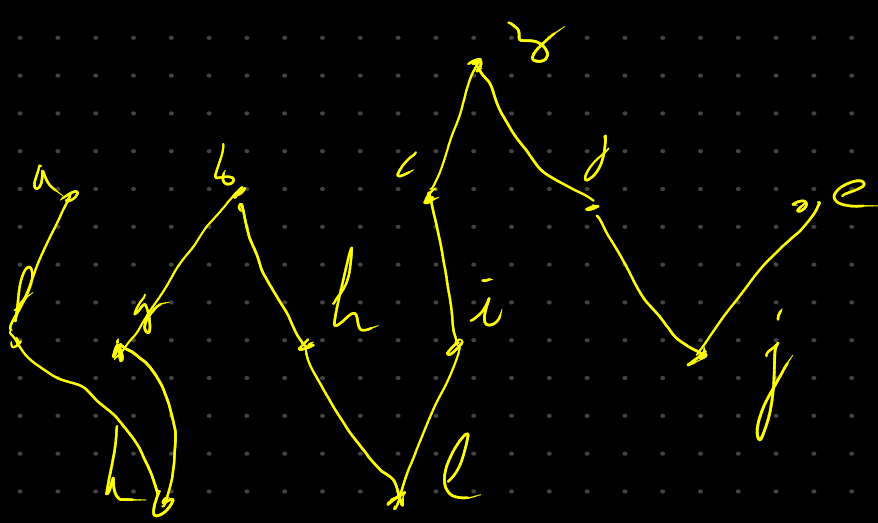




init
 $sync = ($
 $E = \{$
 $adj = \{$
 $ways = \{$
 $PTE = false$
 $firstRel = true$

* $i = 1$, nodes = $\{(k, 1), (l, 2)\}$

$P \neq sync \neq (sync)$

* * * (for loop) $n = (k, 1)$

* * * (for loop) $adj = f$
 $\geq adj$ is not source

* * * (for loop) $adj = g$

* * * (for loop) $n = (l, 2)$

[source]

E is empty

$adj_ways = \{(k, 1)\}$
 $adj_ways = \{(k, 1)\}$

~~***~~ $\text{linksync}(\{ \})$

~~***~~ $\text{sync} f(\{ \})$

$\{f, g, h, i, j\}$

~~***~~ $n = (f, 1)$

~~***~~ $\text{adj} = k$

— adj is source

— $\text{has } IE = \text{true}$

— $E.\text{add}((k, f))$

— $\text{adj source}[k].\text{add}(f)$

$E = [(k, f)]$

$\text{adj source} = \{$

$\quad \quad \quad k : f$

$\}$

~~***~~ $\text{adj} = 1$

not source

~~***~~ $n = g \Rightarrow$

$E = [(k, f), (k, g)]$

$\text{adj} = \{k : [f, g]\}$

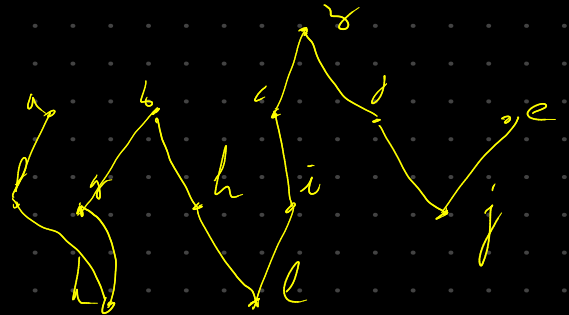
~~***~~ $n = h \Rightarrow$

$E = [(k, f), (k, g), (h, h)]$

$\text{adj} = \{k : [f, g], h : [h]\}$

$E = [(k, f); (k, g); (k, l); (i, l)]$
 $U, S = \{k; [f, g]; l; [l, i]\}$
 $warning = [f]$

$(k, f) \quad (k, g), (k, l), (i, l)$



$source = k$
 $r = f$
 $a = f$
 $b = f$
 $c = f$
 $d = warning = f$

(k, k)
 $(f, 0)$

$(f, i) \quad (g, i)$
 $(k, i) \quad (i, i)$

$(k, k) \quad (f, k)$
 (f, i)
 (g, k)
 (g, i)
 $(l, k) \quad (k, i)$

link-sym

issue with reverse order

$Eq: [(k, k); (k, k); (g, g)]$

$Eq: [(f, g); (k, k)]$

$$\text{Eq. } \left\{ \begin{array}{l} (f', g') = (h, k) = [(h, 0), (k, k)] \\ (f, h) = [(k, 0), (0, 1)] \end{array} \right\}$$

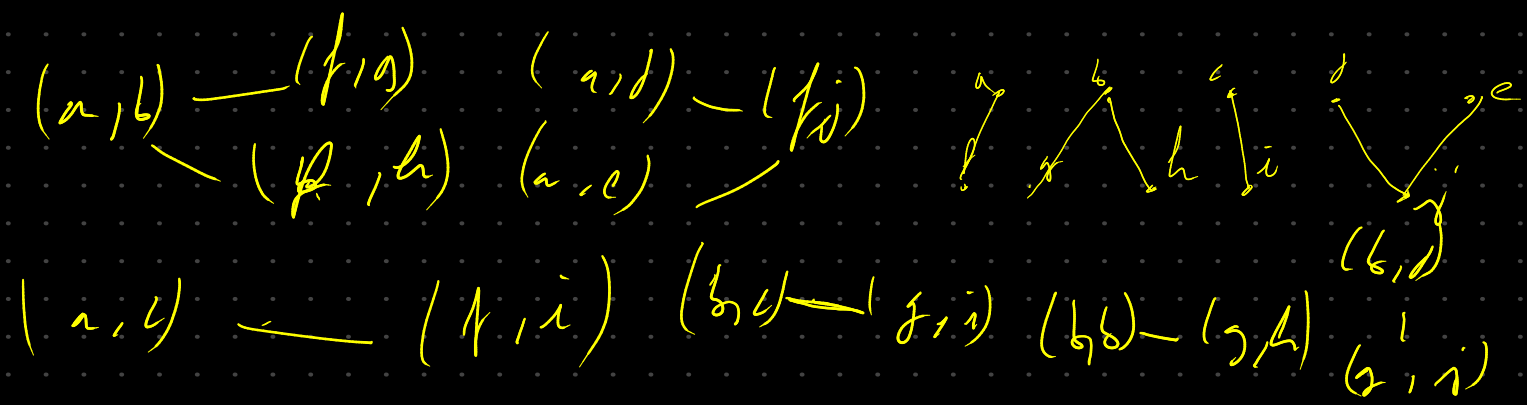
$$\text{Eq. } \left\{ \begin{array}{l} (f', g') = (h, k) = [(h, 0), (k, k)] \\ (k, l) = (f, h) = [(k, 0), (f, h), (f, i); (g, h); (g, i)] \end{array} \right\}$$

$$(f, g) = (h, k)$$

$$\text{Eq. } \begin{array}{l} (k, l) = (f, h) \\ \quad \quad \quad = (f, i) \\ \quad \quad \quad \quad \quad = (g, h) \\ \quad \quad \quad \quad \quad \quad = (g, i) \end{array} \quad \begin{array}{l} (f, i) \quad (g, i) \\ (h, i) \quad (i, i) \end{array}$$

$$(k, l) = (h, i)$$

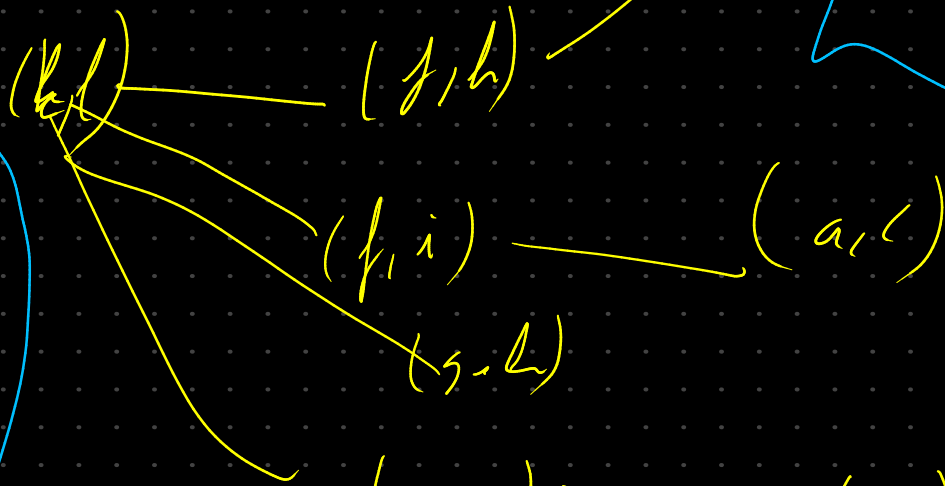
a b c d e



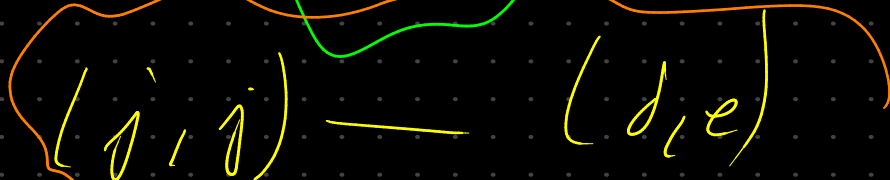
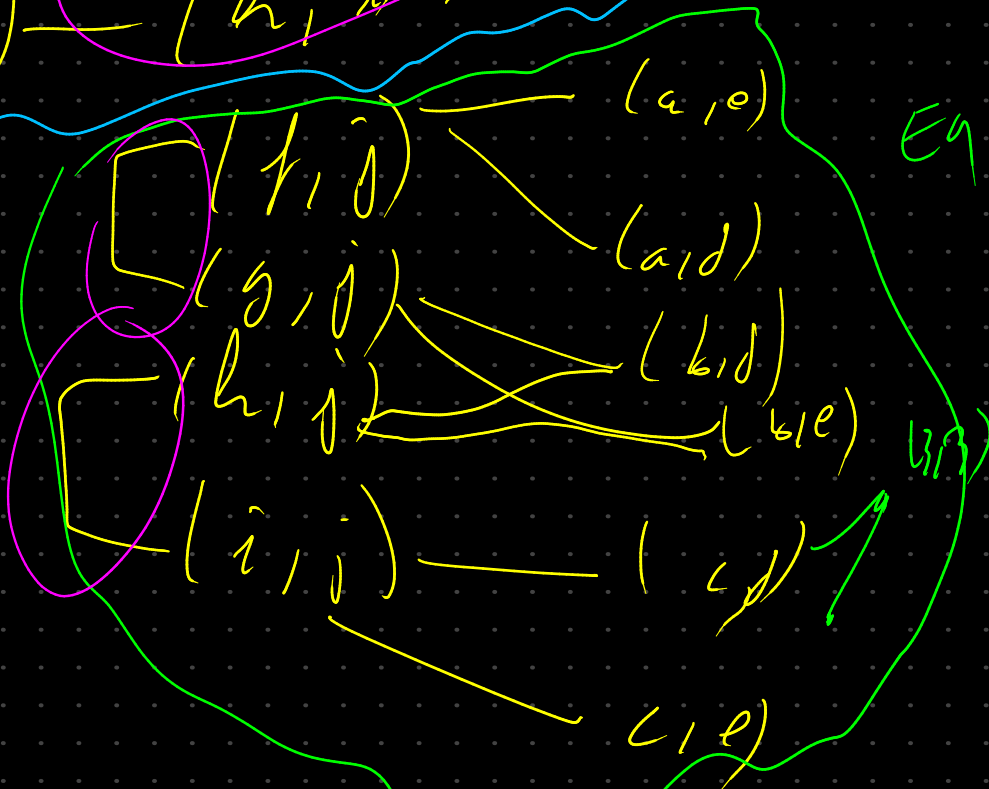
Eq 1

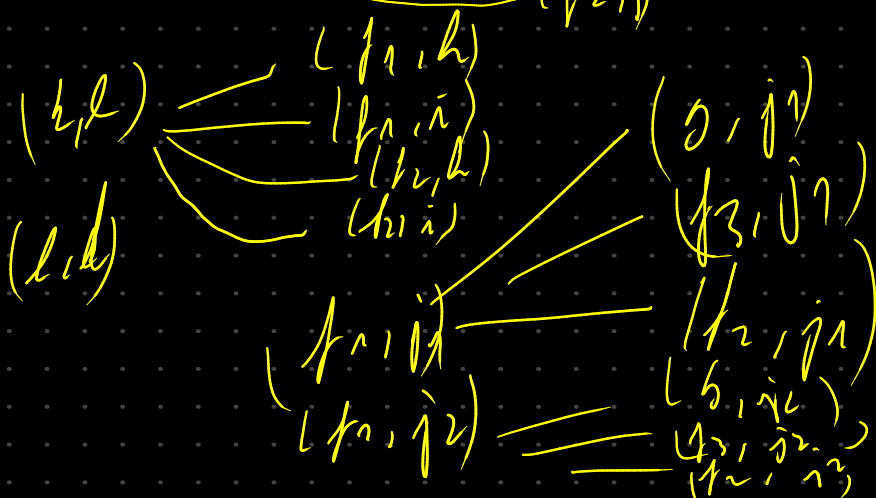
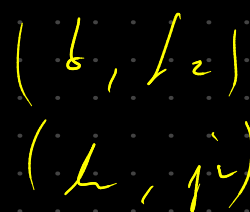
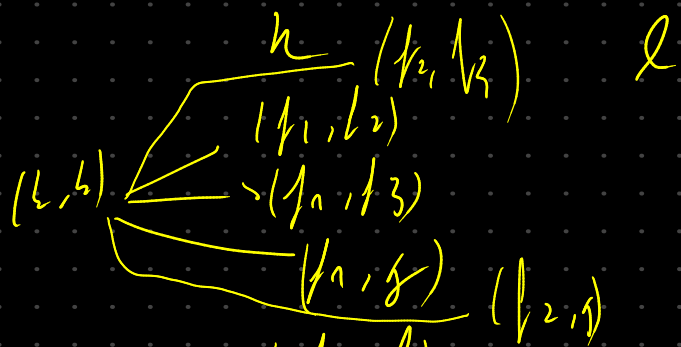
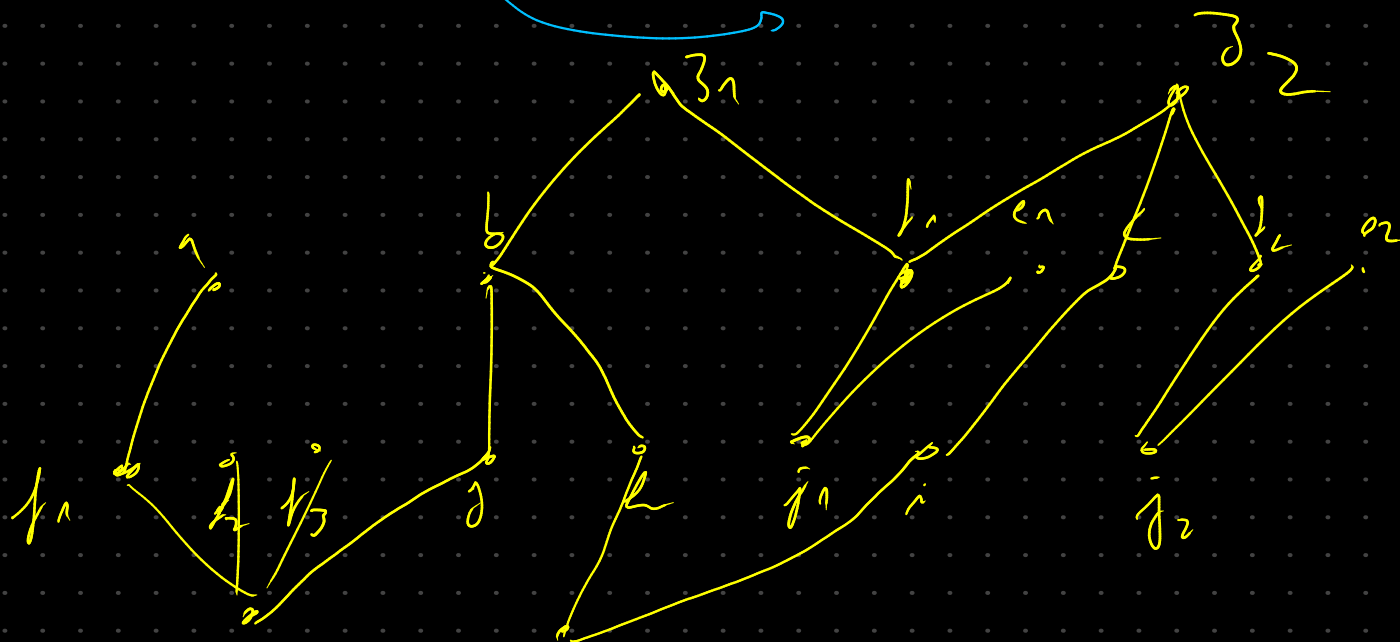
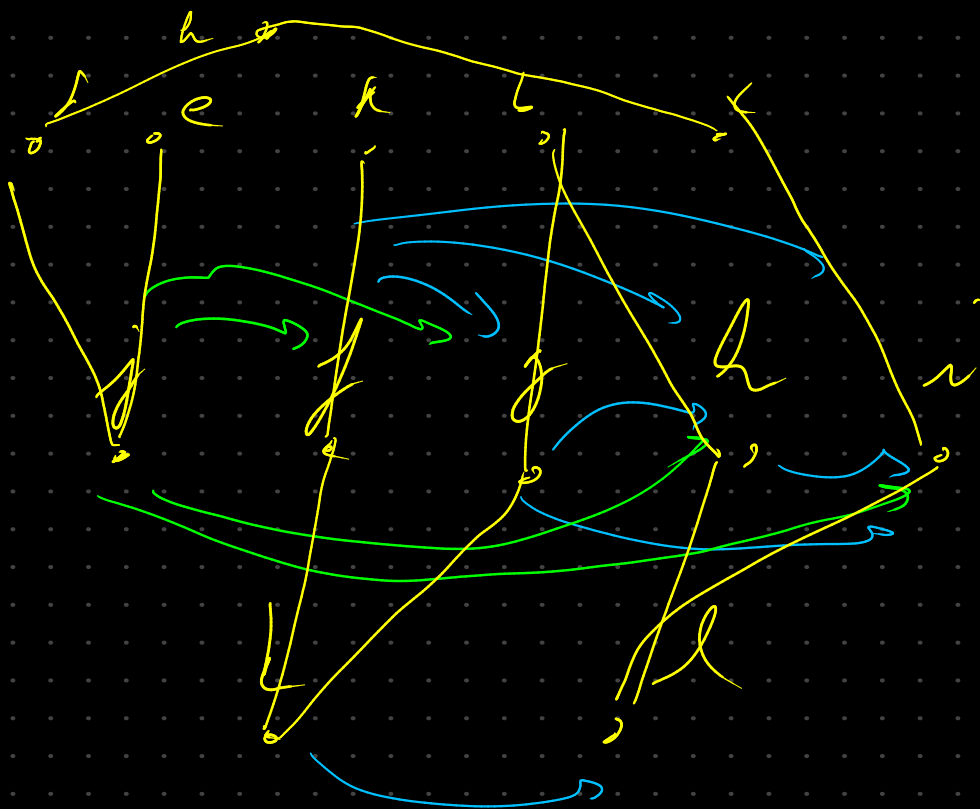


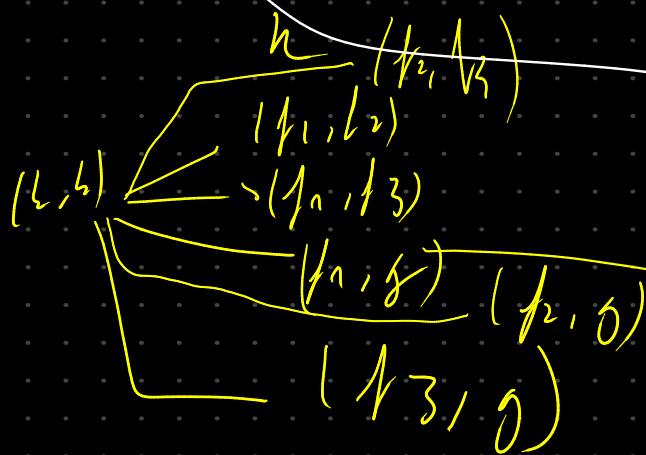
(3,4) (3, 11)



Eq 2







$$(\hat{p}_1, \hat{p}_1) \xrightarrow{\text{E}_1} (d_1, e_1)$$

$$(\hat{p}_1, \hat{p}_2)$$

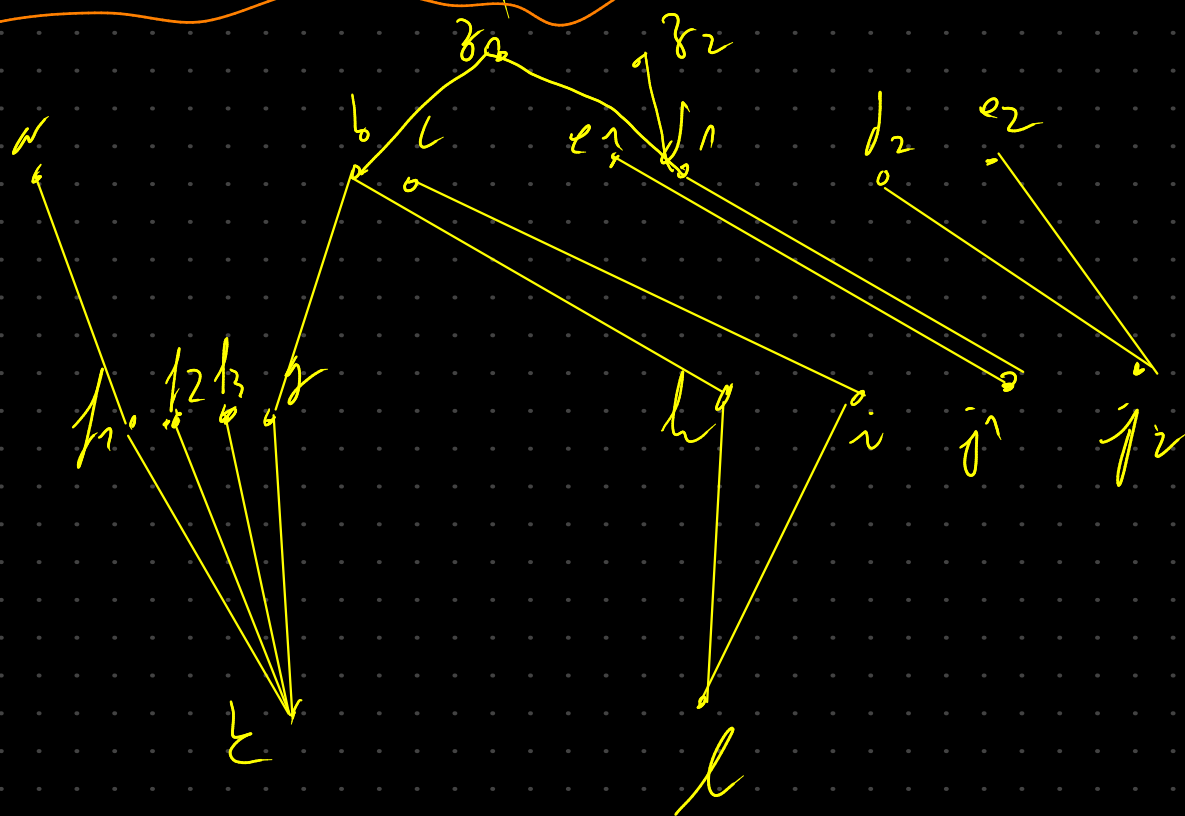
$$(d_1, e_1)$$

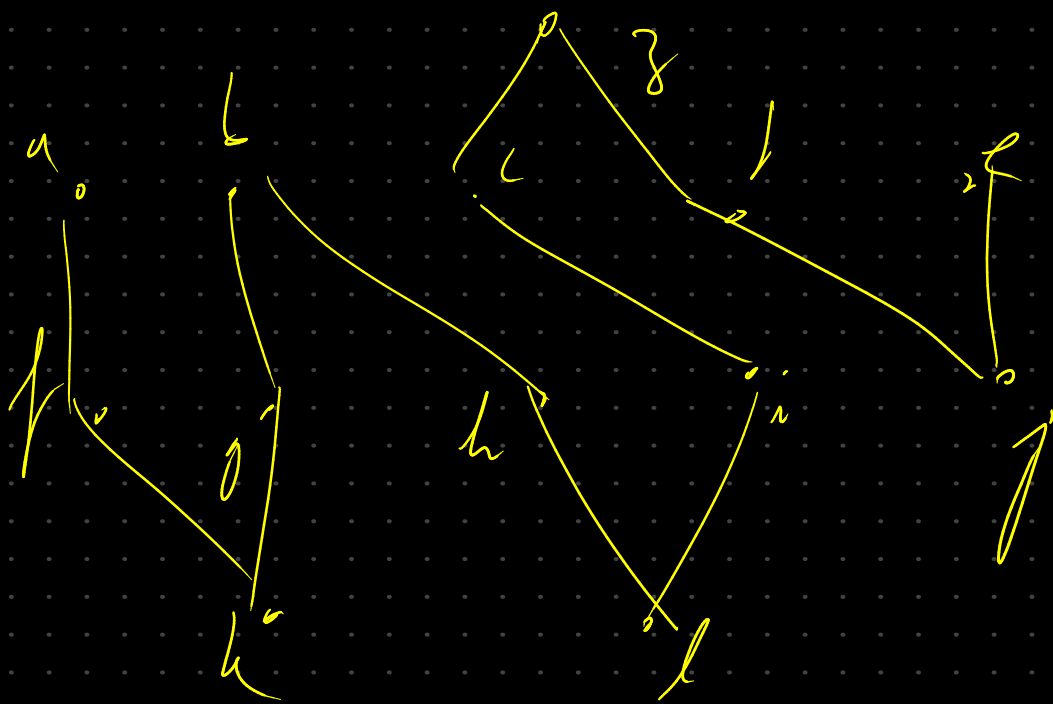
$$(d_1, e_2)$$

$$(e_1, d_2)$$

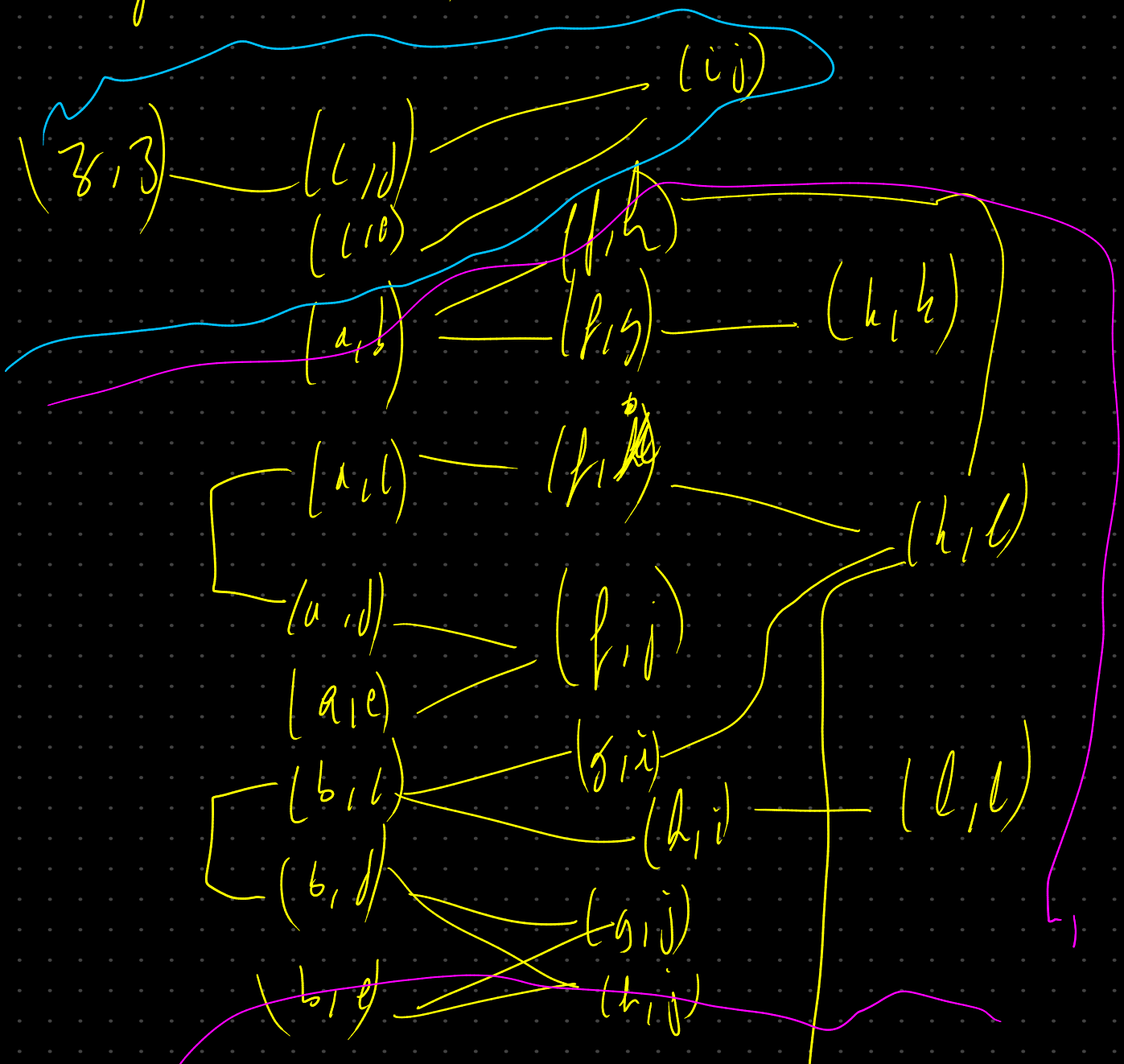
$$(e_1, e_2)$$

$$(\hat{p}_2, \hat{p}_2) \xrightarrow{\text{E}_2} (d_2, e_2)$$





$\text{varnig} = \{a, b, c\}$ $\text{adjacent}[\text{'q'}] = \{c, i\}$

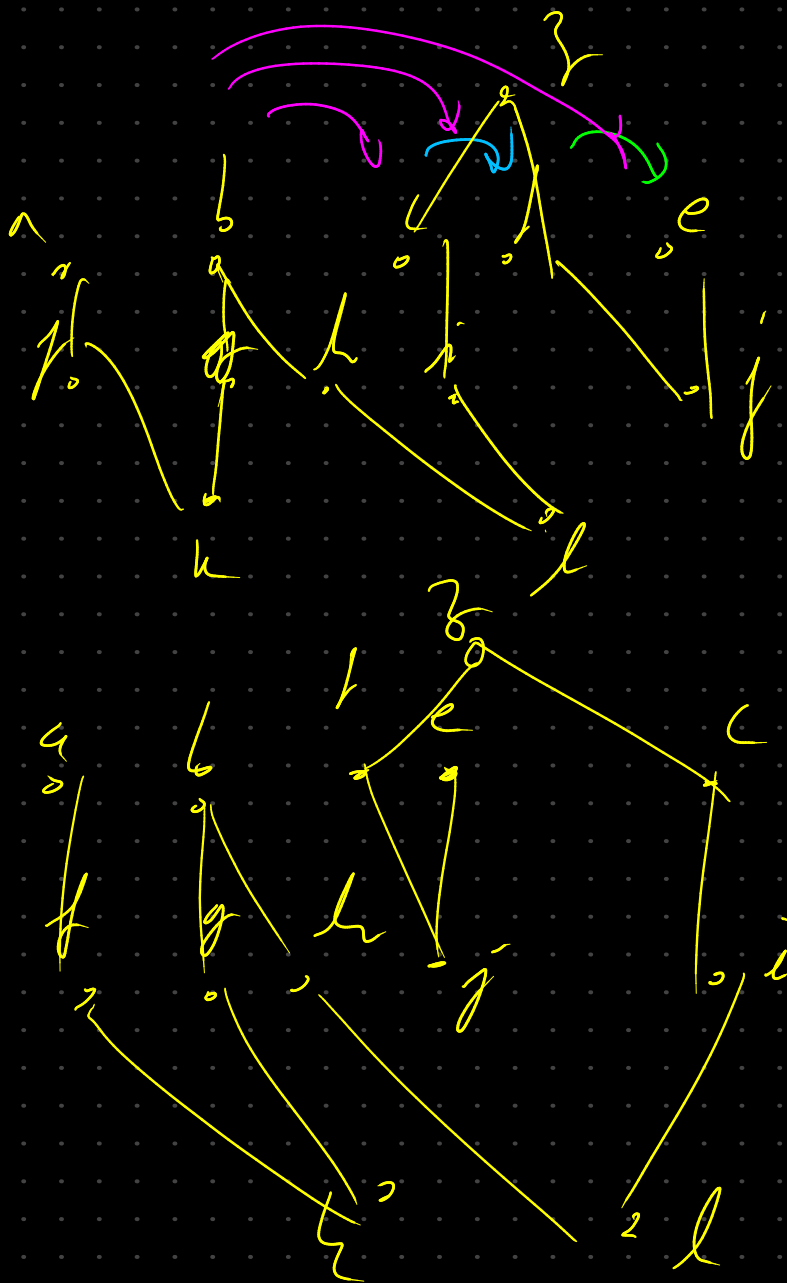


$(i, l) - (j, i)$

$(l, l) - (j, l)$

$E_k \quad E_n \quad E_n$

$\sim \quad \sim \quad \sim$
 $\textcircled{1} \quad \textcircled{1} \quad \textcircled{1}$



12

1, 2, 3, 11

4, 10, 8, 9, 10

5, 6

